SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – PROMPT ENGINEERING CHALLENGE

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 1 – Prompt Engineering
Weightage:	20% of CIA	**Total Marks:**	2 * 50 = 100 (2 Questions)
CO2 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)*		
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Section / Batch:	2024-28	**Date:**	25/10/26

Instructions to Students

• Answer 2 questionss.Each questions carries 50 marks..Total100 Marks.

1. Design appropriate prompts for the given tasks using suitable prompt engineering techniques.
2. Select and apply the appropriate prompting strategy (Zero-shot, One-shot, Few-shot, Chain-of-Thought, or Structured Prompting) based on the given scenario.
3. Include clear instructions, relevant context, necessary constraints, and the expected output format while designing each prompt.
4. Evaluate and refine the designed prompts to improve their clarity, effectiveness, and quality by applying prompt engineering best practices.
5. Design prompts for a variety of real-world applications, such as text generation, summarization, translation, question answering, code generation, and conversational AI.
6. Justify the designed prompt by briefly explaining the selected prompting technique and how it helps achieve the desired output.

Code of Conduct

I _T. Vasudeva Naidu_(192472305)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _T. Vasudeva Naidu_____

1. Design a prompt to explain the Transformer architecture with a suitable diagram description.

1. Aim

To design an effective prompt using Structured Prompting to explain Transformer architecture, its components, and its working process with a suitable diagram description.

2. Prompting Technique

Structured Prompting: This technique organizes the task into clear sections and specifies the required explanation and diagram format.

3. Designed Prompt

Prompt:

You are an expert in Artificial Intelligence and Natural Language Processing. Explain the Transformer architecture in simple English for undergraduate computer science students.
Include the definition, importance, and functions of input embedding, positional encoding, multi-head self-attention, feed-forward networks, encoder, decoder, masked self-attention, encoder-decoder attention, and Softmax output.

Provide a labeled block diagram using boxes and arrows. Explain the data flow step by step, give one real-world application, and list the advantages and limitations. Use clear headings, ensure technical accuracy, distinguish the original encoder-decoder Transformer from decoder-only models, and make sure the diagram matches the explanation.

4. Expected Output

A. Definition

The Transformer is a deep learning architecture introduced in the 2017 research paper "Attention Is All You Need." It uses attention mechanisms to learn relationships between tokens in a sequence and is widely used in Natural Language Processing.

B. Main Components

Input Embedding: Converts input tokens into numerical vectors.

Positional Encoding: Adds information about token positions.

Multi-Head Self-Attention: Identifies relationships between different tokens.

Feed-Forward Network: Processes token representations through neural network layers.

Encoder: Creates contextual representations of the input.

Masked Self-Attention: Prevents the decoder from accessing future output tokens during generation.

Encoder-Decoder Attention: Allows the decoder to use information from the encoder.

Linear and Softmax Layers: Produce probabilities for possible output tokens.

C. Transformer Architecture Diagram

Draw the following labeled blocks and connect them with arrows. The encoder and decoder are shown side by side.

<table>
<tr>
<td>ENCODER

Input Tokens
↓
Input Embedding + Positional Encoding
↓
Multi-Head Self-Attention
↓
Add & Layer Normalization</td>
<td rowspan="2">DECODER

Previous Output Tokens
↓
Embedding + Positional Encoding
↓
Masked Multi-Head Self-Attention
↓
Encoder-Decoder Attention ← Encoder Output</td>
</tr>
<tr>
<td>↓
Feed-Forward Network
↓
Add & Layer Normalization
↓
Encoder Output</td>
</tr>
</table>

Note: The original Transformer repeats encoder and decoder layers. Residual connections and layer normalization surround the attention and feed-forward sublayers. The encoder output is connected to the decoder through encoder-decoder attention.

D. Working Process

The input sentence is divided into tokens.

Input embedding converts tokens into numerical vectors.

Positional encoding provides information about token order.

The encoder uses self-attention to identify relationships between tokens.

The encoder generates contextual representations.

The decoder uses masked self-attention and encoder information to predict output tokens.

The Linear and Softmax layers calculate probabilities for the next token.

The process continues until the output sequence is completed.

E. Real-World Application

Machine Translation: The Transformer can translate a sentence from English into another language. The encoder processes the input sentence, and the decoder generates the translated sentence token by token.

F. Advantages

Supports parallel processing during training.

Captures relationships between tokens.

Works well for several NLP tasks.

Supports machine translation, summarization, and text generation.

G. Limitations

Requires considerable computing resources for large models.

Attention can consume substantial memory for long sequences.

May generate incorrect or misleading information.

5. Evaluation of the Prompt

Clarity: The task and target audience are clearly defined.

Completeness: The main architecture components and working process are specified.

Diagram quality: The prompt requests labeled blocks and directional arrows.

Technical accuracy: It distinguishes the original encoder-decoder Transformer from decoder-only models.

Practical relevance: It includes a real-world application.

6. Prompt Refinement

The initial prompt, “Explain Transformer architecture,” may produce an incomplete answer without a diagram or detailed explanation. The refined prompt specifies the components, working process, diagram structure, application, advantages, limitations, and technical constraints. These additions improve the completeness and accuracy of the response.

7. Justification

Structured Prompting is appropriate because Transformer architecture contains several interconnected components. Clear instructions and a specified diagram format help generate a systematic, technically accurate, and understandable explanation.

8. Conclusion

The Transformer is a fundamental architecture in modern AI. It uses attention mechanisms to process relationships between tokens and supports applications such as translation, summarization, and language generation. A well-designed prompt ensures that its architecture and working process are explained clearly.

2. Write a prompt to compare BERT, GPT, LLaMA, and Gemini in tabular format.

1. Aim

To design an effective prompt using Structured Prompting to compare BERT, GPT, LLaMA, and Gemini based on their features, architecture, applications, advantages, and limitations.

2. Prompting Technique

Structured Prompting: This technique provides clear instructions, relevant context, constraints, and a specific output format.

3. Designed Prompt

Prompt:

You are an expert in Generative AI and Natural Language Processing. Compare BERT, GPT, LLaMA, and Gemini for undergraduate computer science students.

Include the following parameters: developer, architecture, training approach, language understanding, text generation, multimodal capabilities, applications, advantages, and limitations.

Present the answer in a table with the columns Parameter, BERT, GPT, LLaMA, and Gemini. Use simple English, provide technically accurate information, and avoid unsupported claims. After the table, write a short conclusion explaining the suitable applications of each model.

4. Expected Output

Parameter	BERT	GPT	LLaMA	Gemini
Developer	Google	OpenAI	Meta	Google
Architecture	Transformer encoder	Decoder-based Transformer	Decoder-based Transformer	Transformer-based model family
Training Approach	Masked language modeling	Autoregressive next-token prediction	Autoregressive next-token prediction	Varies by model and training stage

Language Understanding	Strong; useful for language analysis tasks	Supports language understanding	Supports language understanding	Supports language understanding across supported modalities
Text Generation	Not its original primary purpose	Yes	Yes	Yes
Multimodal Capabilities	Original BERT is text-focused	Depends on model version	Depends on model version	Available in multimodal versions
Applications	Text classification, sentiment analysis, question answering	Chatbots, coding, summarization, text generation	Research, coding, chatbots, text generation	Image understanding, reasoning, text generation
Advantages	Strong contextual text representations	Flexible text generation	Open-weight options and customization	Broad multimodal capabilities
Limitations	Not designed primarily for open-ended text generation	Capabilities vary by model and version	Computing requirements vary by deployment	Capabilities vary by model and version

Conclusion: BERT is mainly suitable for language understanding, GPT for text generation and conversational AI, LLaMA for customizable AI applications and research, and Gemini for multimodal applications. The best model depends on the task and available resources.

5. Evaluation of the Prompt

Clarity: The task and comparison parameters are clearly specified.

Completeness: The prompt covers architecture, training, capabilities, applications, advantages, and limitations.

Accuracy: It requests technically accurate information and discourages unsupported claims.

Readability: The tabular format makes the comparison easy to follow.

Relevance: The language and content are appropriate for undergraduate students.

6. Prompt Refinement

The initial prompt, “Compare BERT, GPT, LLaMA, and Gemini,” is too general because it does not specify comparison parameters or the output format. The refined prompt adds detailed parameters, the target audience, accuracy requirements, and a table format. This improves clarity, completeness, and consistency.

7. Justification

Structured Prompting is selected because the task requires comparing four AI models across several parameters. Clear instructions and a fixed table format help generate organized, relevant, and easy-to-understand information.

8. Conclusion

A well-designed prompt improves the quality and consistency of AI-generated comparisons. Structured Prompting ensures that the important features of each model are covered systematically.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technique Selection	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Output Effectiveness	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Prompt Craftsmanship	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Prompt-Output Documentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Design Rationale	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness (25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____